Android Application Development

Practical Code Reference Guide

Practical No. 02: Simple "Hello World" program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:gravity="center"
android:orientation="vertical">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Hello World"
android:textSize="28sp" />
</LinearLayout>
```

MainActivity.java

```
package com.example.helloworld;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
}
}
```

Practical No. 03: Programming Resources

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp"
android:gravity="center">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Welcome to Android"
android:textColor="@color/red"
android:textSize="24sp"/>
<ImageView
android:layout_width="150dp"
android:layout_height="150dp"
android:src="@drawable/logo"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Login"/>
</LinearLayout>
```

MainActivity.java (Prac 3)

```
package com.example.aimlprac2;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
}
}
```

colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<color name="blue">#2196F3</color>
<color name="red">#F44336</color>
<color name="white">#FFFFFF</color>
</resources>
```

strings.xml

```
<resources>
<string name="app_name">Resource Demo</string>
<string name="welcome">Welcome to Android</string>
<string name="button_text">Login</string>
</resources>
```

themes.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<style name="Theme.Aimlprac2" parent="Theme.MaterialComponents.DayNight.DarkActionBar">
<item name="colorPrimary">@color/blue</item>
<item name="colorPrimaryVariant">@color/red</item>
<item name="colorOnPrimary">@color/white</item>
</style>
</resources>
```

dimens.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<dimen name="padding">20dp</dimen>
<dimen name="text_size">24sp</dimen>
<dimen name="image_size">150dp</dimen>
</resources>
```

Practical No. 04: Programming Activities and Fragments

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:gravity="center"
android:orientation="vertical">
<TextView
android:layout_width="wrap_content"
```

MainActivity.java (Prac 4)

```
package com.example.activitylifecycledemo;
import android.os.Bundle;
import android.util.Log;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
Toast.makeText(this, "onCreate()", Toast.LENGTH_SHORT).show();
Log.d("ActivityLifeCycle", "onCreate()");
}
@Override
protected void onStart() {
super.onStart();
Toast.makeText(this, "onStart()", Toast.LENGTH_SHORT).show();
Log.d("ActivityLifeCycle", "onStart()");
}
@Override
protected void onResume() {
super.onResume();
Toast.makeText(this, "onResume()", Toast.LENGTH_SHORT).show();
Log.d("ActivityLifeCycle", "onResume()");
}
@Override
protected void onPause() {
super.onPause();
Toast.makeText(this, "onPause()", Toast.LENGTH_SHORT).show();
Log.d("ActivityLifeCycle", "onPause()");
}
@Override
protected void onStop() {
super.onStop();
Toast.makeText(this, "onStop()", Toast.LENGTH_SHORT).show();
Log.d("ActivityLifeCycle", "onStop()");
}
@Override
protected void onRestart() {
super.onRestart();
Toast.makeText(this, "onRestart()", Toast.LENGTH_SHORT).show();
Log.d("ActivityLifeCycle", "onRestart()");
}
@Override
protected void onDestroy() {
super.onDestroy();
Toast.makeText(this, "onDestroy()", Toast.LENGTH_SHORT).show();
Log.d("ActivityLifeCycle", "onDestroy()");
}
}
```

Practical No. 05: Programs related to different Layouts

activity_main.xml

```
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical">
<ListView
android:id="@+id/listView"
android:layout_width="match_parent"
android:layout_height="250dp"/>
<!-- GridView-->
<TextView
android:text="7. GridView"
android:textStyle="bold"
```

MainActivity.java (Prac 5)

```
package com.example.aiml5prc;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
ListView listView;
GridView gridView;
String[] listItems = {
"Apple", "Banana", "Orange", "Mango", "Grapes"
};
String[] gridItems = {
"A", "B", "C", "D", "E", "F"
};
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
listView = findViewById(R.id.listView);
gridView = findViewById(R.id.gridView);
ArrayAdapter<String> listAdapter = new ArrayAdapter<>(
this,
android.R.layout.simple_list_item_1,
listItems);
listView.setAdapter(listAdapter);
ArrayAdapter<String> gridAdapter = new ArrayAdapter<>(
this,
android.R.layout.simple_list_item_1,
gridItems);
gridView.setAdapter(gridAdapter);
}
}
```

Practical No. 06: AppBar, Fragments, UI Components

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical">
<androidx.appcompat.widget.Toolbar
android:id="@+id/toolbar"
android:layout_width="match_parent"
android:layout_height="?attr/actionBarSize"
android:background="#6200EE"
app:title="UI Elements Demo"
app:titleTextColor="@android:color/white"/>
<FrameLayout
android:id="@+id/fragment_container"
android:layout_width="match_parent"
android:layout_height="match_parent"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.aimlprac6;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
public class MainActivity extends AppCompatActivity {
Toolbar toolbar;
@Override
```

fragment_my.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:padding="20dp">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Student Information"
android:textSize="24sp"
android:textStyle="bold"/>
<EditText
android:id="@+id/etName"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Name"/>
<Button
android:id="@+id/btnSubmit"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Submit"/>
<CheckBox
android:id="@+id/check1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Java"/>
<CheckBox
android:id="@+id/check2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Android"/>
<RadioGroup
android:layout_width="wrap_content"
android:layout_height="wrap_content">
<RadioButton
android:id="@+id/r1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Male"/>
<RadioButton
android:id="@+id/r2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Female"/>
</RadioGroup>
<Switch
android:id="@+id/switch1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Enable Notification"/>
</LinearLayout>
</ScrollView>
```

MyFragment.java

```
package com.example.aimlprac6;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
```

Practical No. 07: Menus, Dialog, Dialog Fragments

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:gravity="center"
android:padding="20dp">
<TextView
android:id="@+id/textView"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Menu, Dialog and DialogFragment"
android:textSize="20sp"
android:textStyle="bold" />
<Button
android:id="@+id/btnDialog"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="30dp"
android:text="Show Dialog" />
<Button
android:id="@+id/btnFragment"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="15dp"
android:text="Show Dialog Fragment" />
</LinearLayout>
```

MainActivity.java

```
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AlertDialog;
import android.content.DialogInterface;
import android.view.View;
import android.view.Menu;
import android.view.MenuItem;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
Button btnDialog, btnFragment;
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
btnDialog = findViewById(R.id.btnDialog);
btnFragment = findViewById(R.id.btnFragment);
// Alert Dialog
btnDialog.setOnClickListener(v -> {
new AlertDialog.Builder(MainActivity.this)
.setTitle("Confirmation")
.setMessage("Do you want to continue?")
.setPositiveButton("Yes", (DialogInterface dialog, int which) -> {
Toast.makeText(MainActivity.this, "You clicked Yes", Toast.LENGTH_SHORT).show();
})
.setNegativeButton("No", null)
.show();
});
btnFragment.setOnClickListener(v -> {
// Dialog Fragment
MyDialogFragment dialogFragment = new MyDialogFragment();
dialogFragment.show(getSupportFragmentManager(), "MyDialog");
});
}
// Create Options Menu
@Override
```

MyDialogFragment.java

```
package com.example.aimlprac7;
import android.app.Dialog;
import android.os.Bundle;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AlertDialog;
import androidx.fragment.app.DialogFragment;
public class MyDialogFragment extends DialogFragment {
@NonNull
@Override
public Dialog onCreateDialog(Bundle savedInstanceState) {
return new AlertDialog.Builder(requireContext())
.setTitle("Dialog Fragment")
.setMessage("This is a DialogFragment.")
.setPositiveButton("OK", null)
.setNegativeButton("Cancel", null)
.create();
}
}
```

main_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
<item android:id="@+id/menu_home" android:title="Home" />
<item android:id="@+id/menu_settings" android:title="Settings" />
<item android:id="@+id/menu_exit" android:title="Exit" />
</menu>
```

Practical No. 08: Intents, Events, Listeners and Adapters

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:gravity="center"
android:padding="30dp">
<TextView
android:id="@+id/txtTitle"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Intent, Events and Listeners"
android:textSize="24sp"
android:textStyle="bold"
android:layout_marginBottom="30dp"/>
<Button
android:id="@+id/btnOpen"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Open Second Activity"/>
</LinearLayout>
```

MainActivity.java (Prac 8)

```
package com.example.intenteventadapterdemo;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
Button btnOpen;
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
btnOpen = findViewById(R.id.btnOpen);
// Event Listener
btnOpen.setOnClickListener(v -> {
// Explicit Intent
Intent intent = new Intent(MainActivity.this, SecondActivity.class);
// Start Second Activity
startActivity(intent);
});
}
}
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
<TextView
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Student Subjects"
android:textSize="24sp"
android:textStyle="bold"
android:gravity="center"
android:layout_marginBottom="20dp"/>
<ListView
android:id="@+id/listView"
android:layout_width="match_parent"
android:layout_height="match_parent"/>
</LinearLayout>
```

SecondActivity.java

```
import android.os.Bundle;
import android.widget.ListView;
import android.widget.ArrayAdapter;
import androidx.appcompat.app.AppCompatActivity;
public class SecondActivity extends AppCompatActivity {
ListView listView;
String[] subjects = {
"DBMS", "Operating System", "Android Programming",
"Computer Networks", "Java Programming", "Python"
};
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_second);
listView = findViewById(R.id.listView);
// Adapter
ArrayAdapter<String> adapter = new ArrayAdapter<>(
this,
android.R.layout.simple_list_item_1,
subjects
```

Practical No. 09: Services, Notification and Broadcast Receivers

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:gravity="center"
android:padding="30dp">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Services, Notification and Broadcast"
android:textSize="22sp"
android:textStyle="bold"
android:gravity="center"
android:layout_marginBottom="40dp" />
<Button
android:id="@+id/btnStartService"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="START SERVICE" />
<Button
android:id="@+id/btnStopService"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="STOP SERVICE"
android:layout_marginTop="15dp" />
<Button
android:id="@+id/btnBroadcast"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="SEND BROADCAST"
android:layout_marginTop="15dp" />
</LinearLayout>
```

MainActivity.java (Part 1 - Activity Setup)

```
package com.example.servicenotificationbroadcast;
import android.Manifest;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
private Button btnStartService, btnStopService, btnBroadcast;

@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
if (Build.VERSION.SDK_INT > Build.VERSION_CODES.TIRAMISU) {
if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS) !=
PackageManager.PERMISSION_GRANTED) {
requestPermissions(new String[]{Manifest.permission.POST_NOTIFICATIONS}, 100);
}
}
btnStartService = findViewById(R.id.btnStartService);
btnStopService = findViewById(R.id.btnStopService);
btnBroadcast = findViewById(R.id.btnBroadcast);

btnStartService.setOnClickListener(v -> {
```

MainActivity.java (Part 2 - Inner Classes)

```
import android.app.Notification;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.Service;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.os.IBinder;

// SERVICE CLASS
public static class MyService extends Service {
private static final String CHANNEL_ID = "MY_SERVICE_CHANNEL";
private static final int NOTIFICATION_ID = 1;

@Override
public void onCreate() {
super.onCreate();
createNotificationChannel();
Notification notification;
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
notification = new Notification.Builder(this, CHANNEL_ID)
.setContentTitle("Service Running")
.setContentText("My Android Service is running")
.setSmallIcon(android.R.drawable.ic_dialog_info)
.setOngoing(true).build();
} else {
notification = new Notification.Builder(this)
.setContentTitle("Service Running")
.setContentText("My Android Service is running")
.setSmallIcon(android.R.drawable.ic_dialog_info)
.setOngoing(true).build();
}
startForeground(NOTIFICATION_ID, notification);
}

@Override
public int onStartCommand(Intent intent, int flags, int startId) {
return START_STICKY;
}

@Override
public void onDestroy() {
super.onDestroy();
NotificationManager manager = (NotificationManager) getSystemService(NOTIFICATION_SERVICE);
if (manager != null) manager.cancel(NOTIFICATION_ID);
}

@Override
public IBinder onBind(Intent intent) { return null; }

private void createNotificationChannel() {
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
NotificationChannel channel = new NotificationChannel(
CHANNEL_ID, "My Service Notification", NotificationManager.IMPORTANCE_DEFAULT);
channel.setDescription("Notification for My Service");
NotificationManager manager = getSystemService(NotificationManager.class);
if (manager != null) manager.createNotificationChannel(channel);
}
}
}

// BROADCAST RECEIVER CLASS
public static class MyReceiver extends BroadcastReceiver {
@Override
public void onReceive(Context context, Intent intent) {
if ("MY_CUSTOM_BROADCAST".equals(intent.getAction())) {
Toast.makeText(context, "Broadcast Received Successfully!", Toast.LENGTH_LONG).show();
}
}
}
```

Practical No. 10: Database Programming with SQLite

MainActivity.java (Prac 10)

```
package com.example.sqlitedemo;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
EditText e1, e2, e3;
SQLiteDatabase db;

@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
e1 = findViewById(R.id.e1);
e2 = findViewById(R.id.e2);
e3 = findViewById(R.id.e3);
Button b1 = findViewById(R.id.b1);
Button b2 = findViewById(R.id.b2);
db = openOrCreateDatabase("StudentDB", MODE_PRIVATE, null);
db.execSQL("CREATE TABLE IF NOT EXISTS student(id INTEGER, name TEXT, course TEXT)");

b1.setOnClickListener(v -> {
db.execSQL("INSERT INTO student VALUES(?,?,?)", new Object[]{
e1.getText().toString(), e2.getText().toString(), e3.getText().toString()
});
Toast.makeText(this, "Data Inserted", Toast.LENGTH_SHORT).show();
});

b2.setOnClickListener(v -> {
Cursor c = db.rawQuery("SELECT * FROM student", null);
StringBuilder s = new StringBuilder();
while (c.moveToNext()) {
s.append("ID: ").append(c.getString(0))
.append("\nName: ").append(c.getString(1))
.append("\nCourse: ").append(c.getString(2))
.append("\n\n");
}
c.close();
new AlertDialog.Builder(this)
.setTitle("Student Records")
.setMessage(s.toString())
.setPositiveButton("OK", null)
.show();
});
}
}
```